

INTERFACE CIRCUITS

TYPES SN55325, SN75325 MEMORY CORE DRIVERS

BULLETIN NO. DL-S 7711437, MARCH 1971 — REVISED APRIL 1977

SERIES 55/75 MEMORY DRIVER featuring

PERFORMANCE

- 600-mA Output Capability
- Fast Switching Times
- Output Transient-Voltage Protection
- Dual Sink and Dual Source Outputs
- Minimum Time Skew between Address and Output Current Rise
- 24-Volt Output Capability

EASE OF DESIGN

- Source Base Drive Externally Adjustable
- TTL or DTL Compatibility
- Input Clamping Diodes
- Transformer Coupling Eliminated
- Reliability Increased
- Drive-Line Lengths Reduced
- Use of External Components Minimized

description

The SN55325 and SN75325 are monolithic integrated circuit memory drivers with logic inputs and are designed for use with magnetic memories.

The devices contain two 600-milliampere source switches and two 600-milliampere sink switches. Source selection is determined by one of two logic inputs, and source turn-on is determined by the source strobe. Likewise, sink selection is determined by one of two logic inputs, and sink turn-on is determined by the sink strobe. This arrangement allows selection of one of the four switches and its subsequent turn-on with minimum time skew of the output current rise.

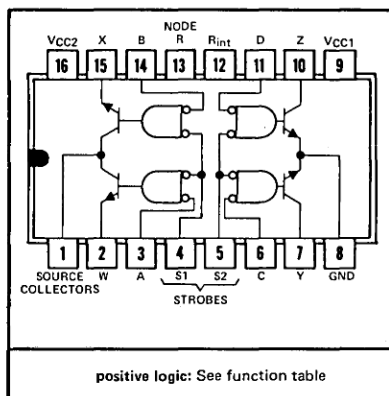
When R_{int} and node R are connected together, the amount of base drive available for the source-1 or source-2 output transistor is set internally by a 575-ohm resistor. This method provides adequate base drive for source currents up to 375 mA with a V_{CC2} voltage of 15 volts or 600 mA with a V_{CC2} voltage of 24 volts.

When source currents greater than 375 mA are required, it is recommended that a resistor of the appropriate value be connected between V_{CC2} and node R and R_{int} must remain open. By using this method the source base current may usually be regulated within $\pm 5\%$. An advantage of this method of setting the base drive is that the power dissipated by this resistor is external to the package and allows the integrated circuit to operate at higher source currents for a given junction temperature.

Each sink-output collector has an internal pull-up resistor in parallel with a clamping diode connected to V_{CC2} . This arrangement provides protection from voltage surges associated with switching inductive loads.

The SN55325 is characterized for operation over the full military temperature range of -55°C to 125°C ; the SN75325 is characterized for operation from 0°C to 70°C .

J OR N DUAL-IN-LINE PACKAGE
(TOP VIEW)



positive logic: See function table

FUNCTION TABLE

ADDRESS INPUTS				STROBE INPUTS		OUTPUTS			
SOURCE	SINK	SOURCE	SINK	SOURCE	SINK	SOURCE	SINK	SOURCE	SINK
A	B	C	D	S1	S2	W	X	Y	Z
L	H	X	X	L	H	ON	OFF	OFF	OFF
H	L	X	X	L	H	OFF	ON	OFF	OFF
X	X	L	H	H	L	OFF	OFF	ON	OFF
X	X	H	L	H	L	OFF	OFF	OFF	ON
X	X	X	X	H	H	OFF	OFF	OFF	OFF
H	H	H	H	X	X	OFF	OFF	OFF	OFF

H = high level, L = low level, X = irrelevant

NOTE: Not more than one output is to be on at any one time.

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absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

	SN55325	SN75325	UNIT
Supply voltage V_{CC1} (see Note 1)	7	7	V
Supply voltage V_{CC2} (see Note 1)	25	25	V
Input voltage (any address or strobe input)	5.5	5.5	V
Continuous total dissipation at (or below) 25°C free-air temperature (see Note 2)	J package	1375	mW
	N package	1150	
Operating free-air temperature range	-55 to 125	0 to 70	°C
Storage temperature range	-65 to 150	-65 to 150	°C
Lead temperature 1/16 inch from case for 60 seconds	J package	300	°C
Lead temperature 1/16 inch from case for 10 seconds	N package	260	°C

NOTES: 1. Voltage values are with respect to network ground terminal.

2. For operation above 25°C free-air temperature, refer to Dissipation Derating Curves in the Thermal Information section, which starts on page 19. In the J package, SN55325 and SN75325 chips are alloy-mounted.

electrical characteristics over rated operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST FIGURE	TEST CONDITIONS	SN55325		SN75325		UNIT
				MIN	TYP† MAX	MIN	TYP† MAX	
V_{IH}	High-level input voltage	1 & 2		2		2		V
V_{IL}	Low-level input voltage	3 & 4			0.8		0.8	V
V_{IK}	Input clamp voltage	5	$V_{CC1} = 4.5$ V, $I_I = -10$ mA, $T_A = 25^\circ$ C		-1.3 -1.7		-1.3 -1.7	V
$I_{(off)}$	Source-collectors terminal off-state current	1	$V_{CC1} = 4.5$ V, $V_{CC2} = 24$ V, $T_A = 25^\circ$ C		500		200	μA
V_{OH}	High-level sink output voltage	2	$V_{CC1} = 4.5$ V, $I_O = 0$, $V_{CC2} = 24$ V	19	23	19	23	V
$V_{(sat)}$	Saturation voltage	3	$V_{CC1} = 4.5$ V, $V_{CC2} = 15$ V, $R_L = 24 \Omega$, $I_{(source)} \approx -600$ mA‡, See Note 3		$T_A = \text{full range } \dagger$ 0.9		$T_A = 25^\circ$ C 0.9	V
					0.43 0.7		0.43 0.75	
	sink outputs	4	$V_{CC1} = 4.5$ V, $V_{CC2} = 15$ V, $R_L = 24 \Omega$, $I_{(sink)} \approx 600$ mA‡, See Note 3		$T_A = \text{full range } \dagger$ 0.9		$T_A = 25^\circ$ C 0.9	
					0.43 0.7		0.43 0.75	
I_I	Input current at maximum input voltage	5	$V_{CC1} = 5.5$ V, $V_I = 5.5$ V, $V_{CC2} = 24$ V		1		1	mA
					2		2	
I_{IH}	High-level input current	5	$V_{CC1} = 5.5$ V, $V_I = 2.4$ V, $V_{CC2} = 24$ V		3 40		3 40	μA
I_{IL}	Low-level input current	5	$V_{CC1} = 5.5$ V, $V_I = 0.4$ V, $V_{CC2} = 24$ V		-1 -1.6		-1 -1.6	mA
$I_{CC(off)}$	Supply current, all sources and sinks off	6	$V_{CC1} = 5.5$ V, $T_A = 25^\circ$ C, $V_{CC2} = 24$ V		14 22		14 22	mA
					7.5 20		7.5 20	
I_{CC1}	Supply current from V_{CC1} , either sink on	7	$V_{CC1} = 5.5$ V, $I_{(sink)} = 50$ mA, $T_A = 25^\circ$ C, $V_{CC2} = 24$ V		55 70		55 70	mA
I_{CC2}	Supply current from V_{CC2} , either source on	8	$V_{CC1} = 5.5$ V, $I_{(source)} = -50$ mA, $T_A = 25^\circ$ C, See Note 3		32 50		32 50	mA

† Full range for SN55325 is -55°C to 125°C and for SN75325 is 0°C to 70°C.

‡ All typical values are at $T_A = 25^\circ$ C.

§ Under these conditions, not more than one output is to be on at any one time.

NOTE 3: These parameters must be measured using pulse techniques, $t_W = 200 \mu$ s, duty cycle $\leq 2\%$.

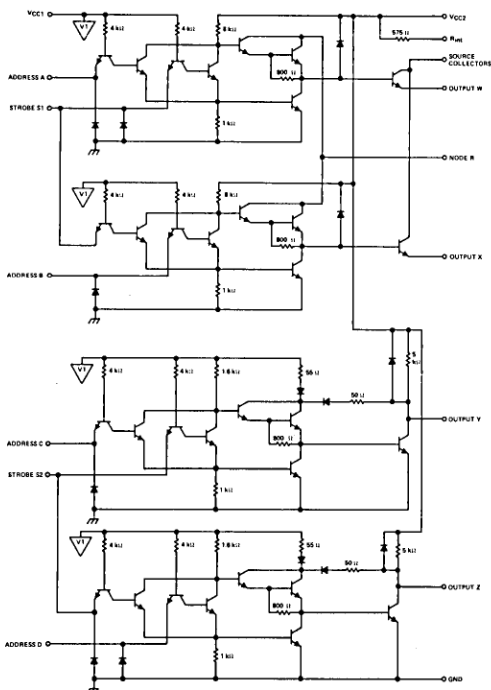
TYPES SN55325, SN75325 MEMORY CORE DRIVERS

switching characteristics, $V_{CC1} = 5\text{ V}$, $T_A = 25^\circ\text{C}$

PARAMETER [†]	TO (OUTPUT)	TEST FIGURE	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t _{PLH}	Source collectors	9	V _{CC2} = 15 V, R _L = 24 Ω, C _L = 25 pF	35	50		ns
t _{PHL}				35	50		
t _{TLH}	Source outputs	10	V _{CC2} = 20 V, R _L = 1 kΩ, C _L = 25 pF	55			ns
t _{THL}				7			
t _{PLH}	Sink outputs	9	V _{CC2} = 15 V, R _L = 24 Ω, C _L = 25 pF	20	45		ns
t _{PHL}				20	45		
t _{TLH}	Sink outputs	9	V _{CC2} = 15 V, R _L = 24 Ω, C _L = 25 pF	7	15		ns
t _{THL}				9	20		
t _s	Sink outputs	9	V _{CC2} = 15 V, R _L = 24 Ω, C _L = 25 pF	15	30		ns

- t_{PLH} = propagation delay time, low-to-high-level output
- t_{PHL} = propagation delay time, high-to-low-level output
- t_{TLH} = transition time, low-to-high-level output
- t_{THL} = transition time, high-to-low-level output
- t_s = storage time

schematic



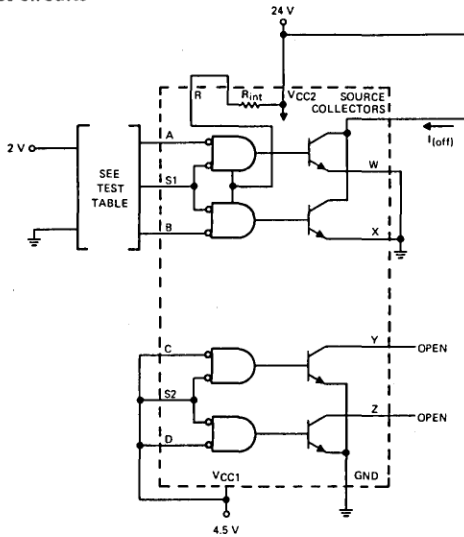
Component values shown are nominal.

V1 ... **V_{CC1}** bus

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d-c test circuits†

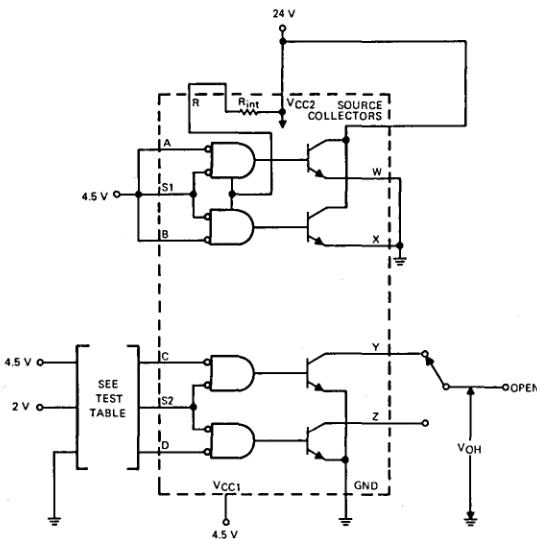
PARAMETER MEASUREMENT INFORMATION



TEST TABLE

A	B	S1
GND	GND	2 V
2 V	2 V	GND

FIGURE 1— V_{IH} AND $I_{(off)}$



TEST TABLE

C	D	S2	Y	Z
2 V	4.5 V	GND	V_{OH}	OPEN
GND	4.5 V	2 V	V_{OH}	OPEN
4.5 V	2 V	GND	OPEN	V_{OH}
4.5 V	GND	2 V	OPEN	V_{OH}

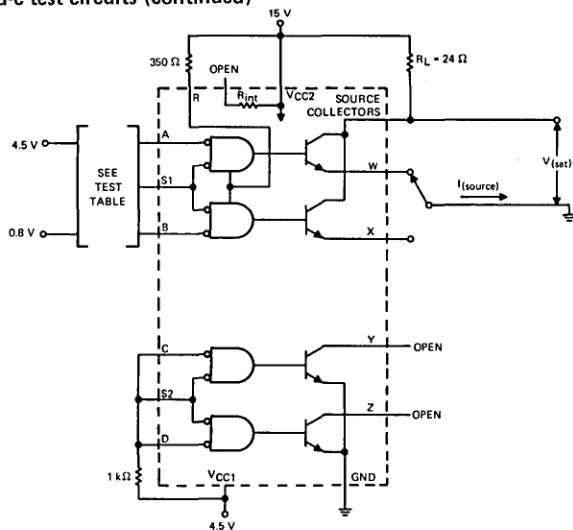
FIGURE 2— V_{IH} AND V_{OH}

†Arrows indicate actual direction of current flow.

TYPES SN55325, SN75325 MEMORY CORE DRIVERS

PARAMETER MEASUREMENT INFORMATION

d-c test circuits (continued)[†]

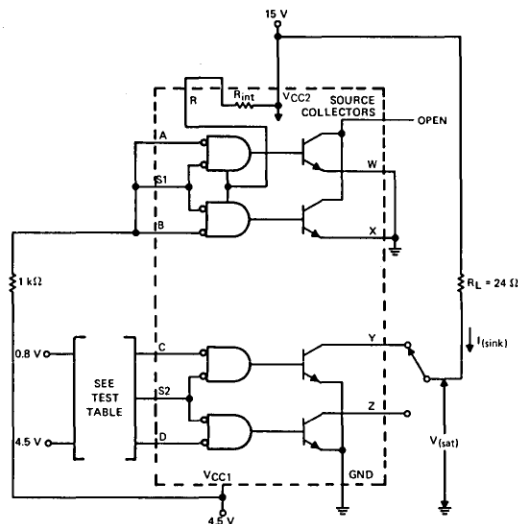


TEST TABLE

A	B	S1	W	X
0.8 V	4.5 V	0.8 V	GND	OPEN
4.5 V	0.8 V	0.8 V	OPEN	GND

NOTE A: These parameters must be measured using pulse techniques. $t_w = 200 \mu s$, duty cycle $\leq 2\%$.

FIGURE 3— V_{IL} AND SOURCE $V_{(sat)}$



TEST TABLE

C	D	S2	Y	Z
0.8 V	4.5 V	0.8 V	R_L	OPEN
4.5 V	0.8 V	0.8 V	OPEN	R_L

NOTE A: These parameters must be measured using pulse techniques. $t_w = 200 \mu s$, duty cycle $\leq 2\%$.

FIGURE 4— V_{IL} AND SINK $V_{(sat)}$

[†] Arrows indicate actual direction of current flow.

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PARAMETER MEASUREMENT INFORMATION

d-c test circuits (continued)[†]

TEST TABLES

I_I, I_{IH}

APPLY $V_I = 5.5$ V, MEASURE I_I	GROUND	APPLY 5.5 V
APPLY $V_I = 2.4$ V, MEASURE I_{IH}		
A	S1	B, C, S2, D
S1	A, B	C, S2, D
B	S1	A, C, S2, D
C	S2	A, S1, B, D
S2	C, D	A, S1, B
D	S2	A, S1, B, C

V_I, I_{IL}

APPLY $V_I = 0.4$ V, MEASURE I_{IL}	APPLY 5.5 V
APPLY $I_I = -10$ mA MEASURE V_I	
A	S1, B, C, S2, D
S1	A, B, C, S2, D
B	A, S1, C, S2, D
C	A, S1, B, S2, D
S2	A, S1, B, C, D
D	A, S1, B, C, S2

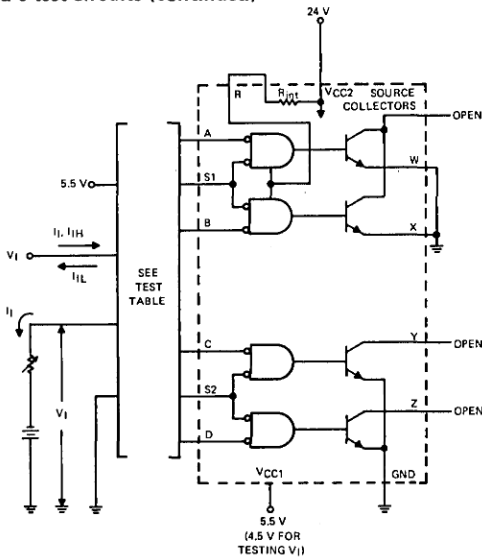


FIGURE 5— V_I, I_I, I_{IH} , AND I_{IL}

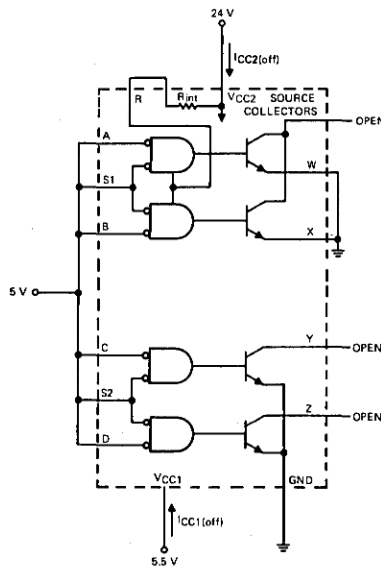


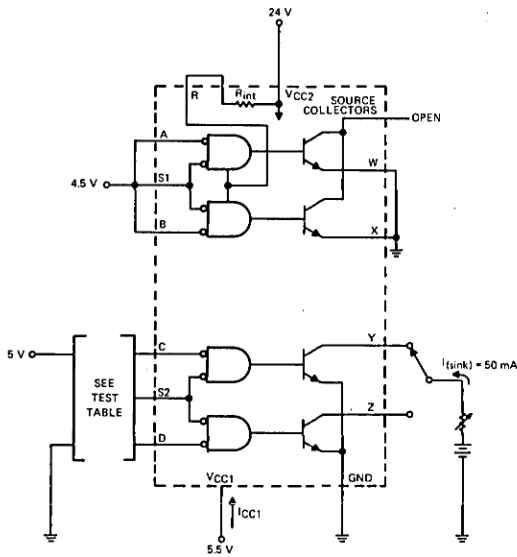
FIGURE 6— $I_{CC1}(\text{off})$ AND $I_{CC2}(\text{off})$

[†] Arrows indicate actual direction of current flow.

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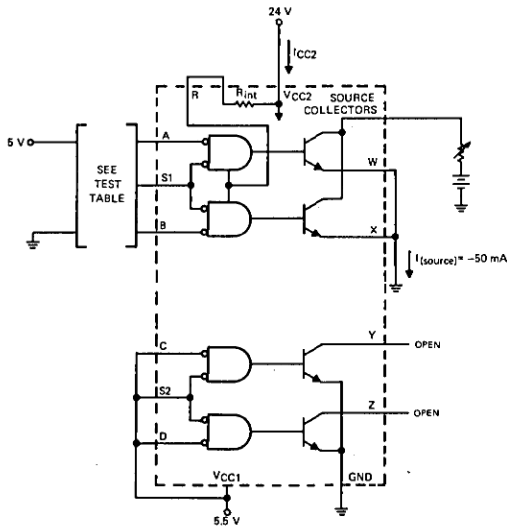
PARAMETER MEASUREMENT INFORMATION

d-c test circuits (continued)[†]



TEST TABLE				
C	D	S2	Y	Z
GND	5 V	GND	$I_{(sink)}$	OPEN
5 V	GND	GND	OPEN	$I_{(sink)}$

FIGURE 7— I_{CC1} , EITHER SINK ON



TEST TABLE		
A	B	S1
GND	5 V	GND
5 V	GND	GND

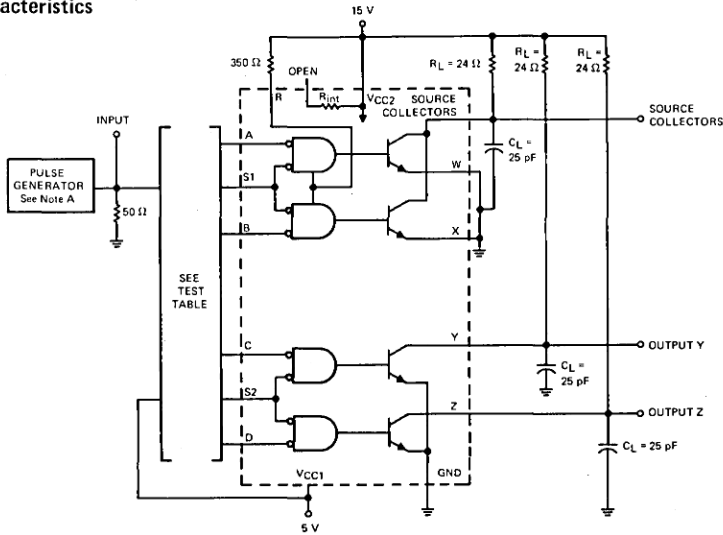
FIGURE 8— I_{CC2} , EITHER SOURCE ON

[†]Arrows indicate actual direction of current flow.

TYPES SN55325, SN75325 MEMORY CORE DRIVERS

PARAMETER MEASUREMENT INFORMATION

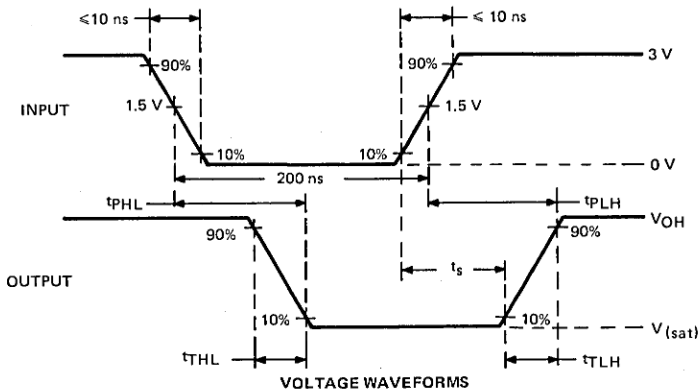
switching characteristics



TEST CIRCUIT

TEST TABLE

PARAMETER	OUTPUT UNDER TEST	INPUT	CONNECT TO 5 V
t_{PLH} and t_{PHL}	Source collectors	A and S1 B and S1	B, C, D and S2 A, C, D and S2
t_{PLH} , t_{PHL} , t_{TLH} , t_{THL} , and t_s	Sink output Y	C and S2	A, B, D and S1
	Sink output Z	D and S2	A, B, C and S1



VOLTAGE WAVEFORMS

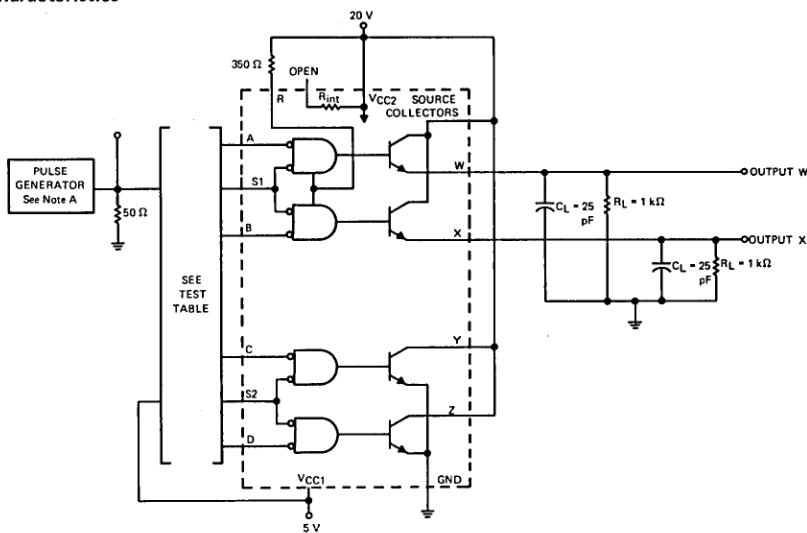
NOTES: A. The pulse generator has the following characteristics: $Z_{out} = 50 \Omega$, duty cycle $\leq 1\%$.
B. C_L includes probe and jig capacitance.

FIGURE 9—SWITCHING TIMES

TYPES SN55325, SN75325 MEMORY CORE DRIVERS

PARAMETER MEASUREMENT INFORMATION

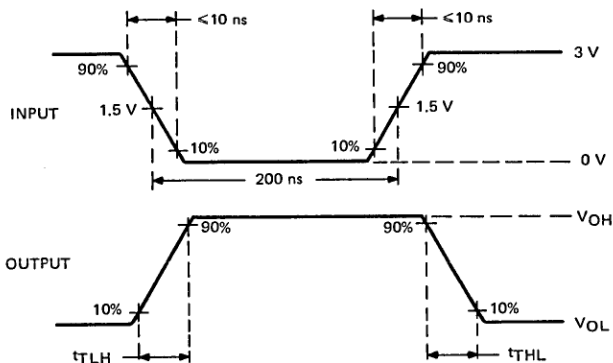
switching characteristics



TEST CIRCUIT

TEST TABLE

PARAMETER	OUTPUT UNDER TEST	INPUT	CONNECT TO 5 V
t_{TLH} and t_{THL}	Source output W	A and S1	B, C, D, and S2
	Source output X	B and S1	A, C, D, and S2



VOLTAGE WAVEFORMS

- NOTES: A. The pulse generator has the following characteristics: $Z_{out} = 50 \Omega$, duty cycle $\leq 1\%$.
B. C_L includes probe and jig capacitance.

FIGURE 10—TRANSITION TIMES OF SOURCE OUTPUTS

TYPES SN55325, SN75325 MEMORY CORE DRIVERS

TYPICAL CHARACTERISTICS

OFF-STATE CURRENT INTO SOURCE COLLECTORS

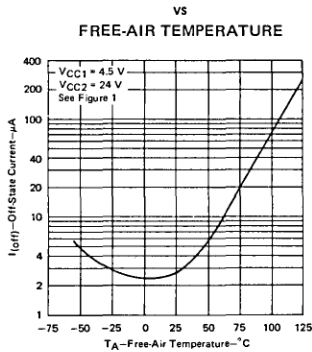


FIGURE 11

HIGH-LEVEL SINK OUTPUT VOLTAGE

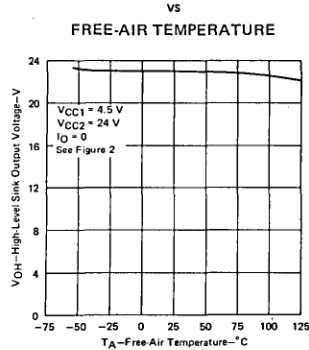


FIGURE 12

SOURCE OR SINK SATURATION VOLTAGE

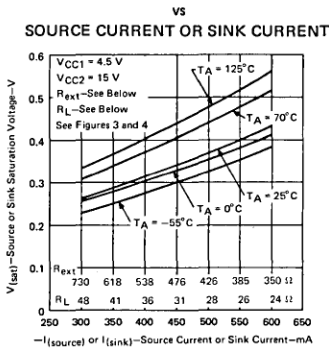


FIGURE 13

SOURCE OR SINK SATURATION VOLTAGE

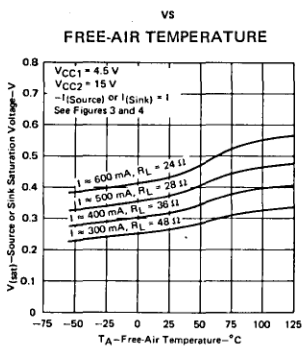


FIGURE 14

SUPPLY CURRENT, ALL SOURCES AND SINKS OFF

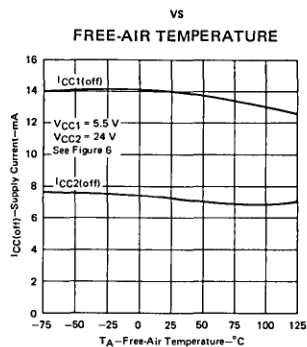


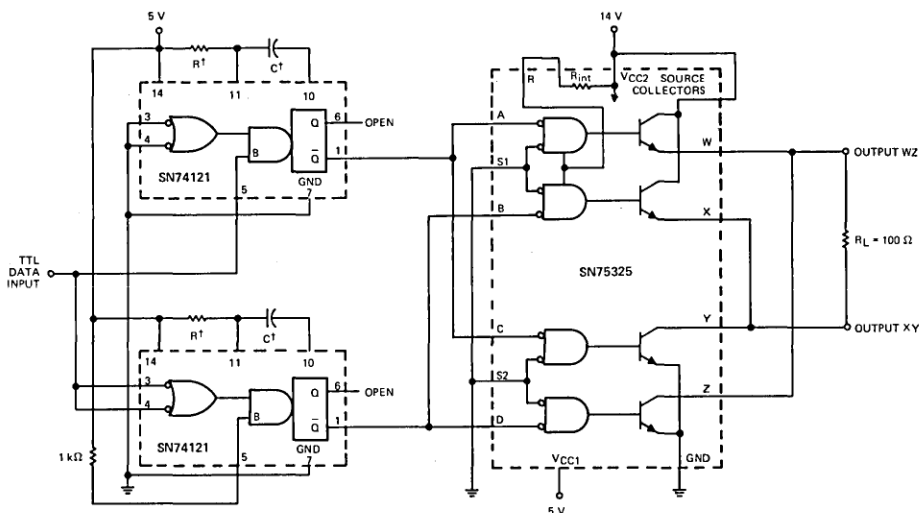
FIGURE 15

TYPES SN55325, SN75325 MEMORY CORE DRIVERS

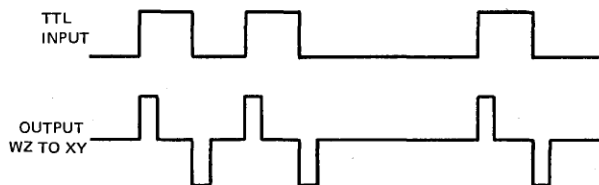
TYPICAL APPLICATION DATA

balanced bipolar logic-line driver

The circuit shown in Figure 16 converts standard TTL logic to bipolar logic. Bipolar logic is primarily used in transmitting data or clock pulses over long lines. This line-driver may be operated from a single 5-volt supply; however, the output drive may be increased by raising the supply voltage to the source collectors. The circuit features a three-state output which is off during the absence of data, thus not dissipating high power. It provides a balanced drive circuit giving maximum noise immunity when used with the proper line receiver. Large drive levels can be used to further increase noise immunity. The circuit is capable of driving twisted-pair lines of several thousand feet in length or low-impedance coaxial lines.



TEST CIRCUIT



VOLTAGE WAVEFORMS

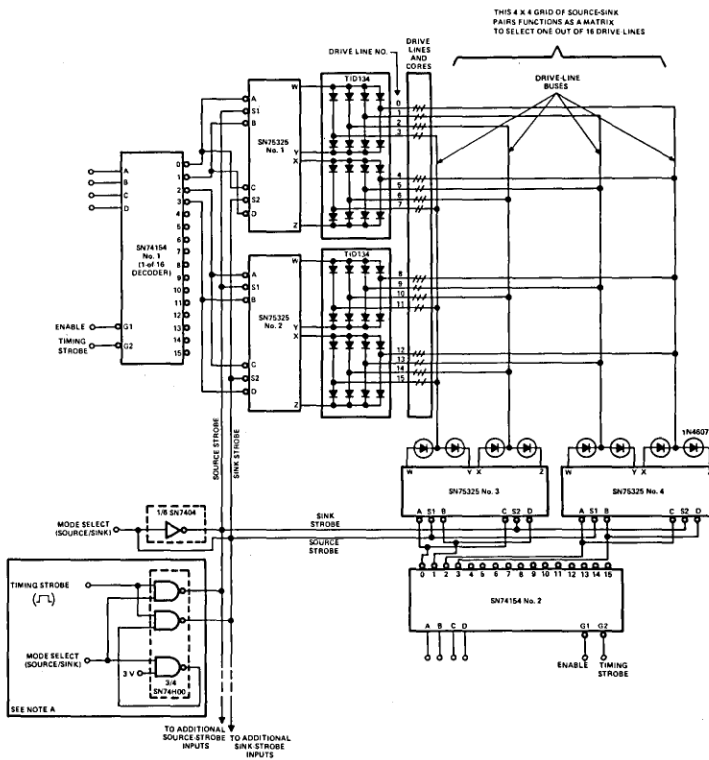
† R and C are adjusted to give the desired bipolar output pulse width.

FIGURE 16—BALANCED BIPOLAR LOGIC—LINE DRIVER

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TYPICAL APPLICATION DATA

In memory-drive applications the SN75325 (or for full-temperature operation, the SN55325) can be connected in any of several ways. Typically, however, sources and sinks are arranged in pairs from which many drive-lines branch off as shown in Figure 17. Here each drive-line is served by a unique combination of two source/sink pairs so that a selection matrix is formed. To select drive-line 13, SN74154 No. 1 must be set to 3 (with mode select high), enabling source X of SN75325 No. 2 to drive lines 12 through 15, and SN74154 No. 2 must be set to 2, providing a sink at Y of SN75325 No. 4 for drive-line 13 only. Alternatively, to drive current in drive-line 13 in the opposite direction, only the mode-select voltage would be changed from high to low. The size of such a matrix is limited only by the number of drive-lines that a source/sink pair can serve. This number in turn depends on the capacitive and inductive load that each drive-line of the particular system imposes on the driver. A 256-drive-line selection matrix is shown in Figure 18. These 256 drive-lines are sufficient to serve $(256/2)^2 = 16,384$ individual cores.



NOTE A: This optional mode-select and timing-strobe technique can be used in place of the SN7440 mode-select and SN74154 timing-strobe when minimum time skew is desired.

FIGURE 17—SN75325 USED AS A MEMORY DRIVER
TO SELECT ONE OF SIXTEEN DRIVE LINES

6

210

37

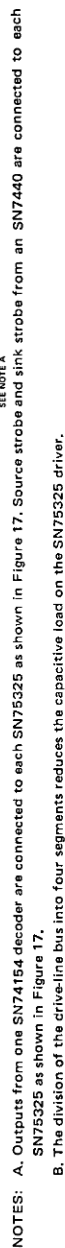


FIGURE 18—SN75325 SERVING 256 DRIVE LINES IN A MAGNETIC MEMORY

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TYPICAL APPLICATION DATA

external resistor calculation

A typical magnetic-memory word-drive requirement is shown in Figure 19. A source-output transistor of one SN75325 delivers load current (I_L). The sink-output transistor of another SN75325 sinks this current.

The value of the external pull-up resistor (R_{ext}) for a particular memory application may be determined using the following equation:

$$R_{ext} = \frac{16 [V_{CC2(min)} - V_S - 2.2]}{I_L - 1.6 [V_{CC2(min)} - V_S - 2.9]} \quad (\text{Equation 1})$$

where: R_{ext} is in $k\Omega$,
 $V_{CC2(min)}$ is the lowest expected value of V_{CC2} in volts,
 V_S is the source output voltage in volts with respect to ground,
 I_L is in mA.

The power dissipated in resistor R_{ext} during the load current pulse duration is calculated using Equation 2,

$$P_{R_{ext}} \approx \frac{I_L}{16} [V_{CC2(min)} - V_S - 2] \quad (\text{Equation 2})$$

where: $P_{R_{ext}}$ is in mW.

After solving for R_{ext} , the magnitude of the source collector current (I_{CS}) is determined from Equation 3,

$$I_{CS} \approx 0.94 I_L \quad (\text{Equation 3})$$

where: I_{CS} is in mA.

As an example, let $V_{CC2(min)} = 20$ V and $V_L = 3$ V while I_L of 500 mA flows.

Using Equation 1,

$$R_{ext} = \frac{16 (20 - 3 - 2.2)}{500 - 1.6 (20 - 3 - 2.9)} = 0.5 \text{ k}\Omega$$

and from Equation 2,

$$P_{R_{ext}} \approx \frac{500}{16} [20 - 3 - 2] \approx 470 \text{ mW}$$

The amount of the memory system current source (I_{CS}) from Equation 3 is:

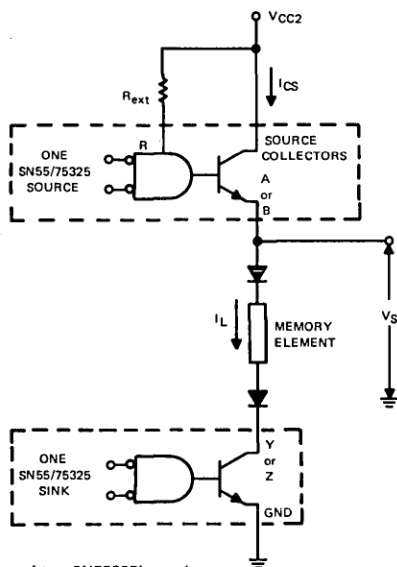
$$I_{CS} \approx 0.94 (500) \approx 470 \text{ mA}$$

In this example the regulated source-output transistor base current through the external pull-up resistor (R_{ext}) and the source gate is approximately 30 mA. This current and I_{CS} comprise I_L .

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TYPICAL APPLICATION DATA

external resistor calculation (continued)



NOTES: A. For clarity, partial logic diagrams of two SN75325's are shown.
B. Source and sink shown are in different packages.

FIGURE 19